Module I – Networking Principles and Layered Architecture

Data Communication & Networking: A Communication Models – Data Communication – Evolution of network, Requirements, Applications, Network Topology, Protocols and Standards, Network Models(OSI and TCP/IP)

DATA COMMUNICATIONS

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Topics discussed in this section:

Components
Data Représentation
Data Flow

Effectiveness of Data communication

Should depend on 4 fundamental characteristics

- Delivery
- Accuracy
- Timeliness
- Jitter

Figure 1.1 *Five components of data communication*

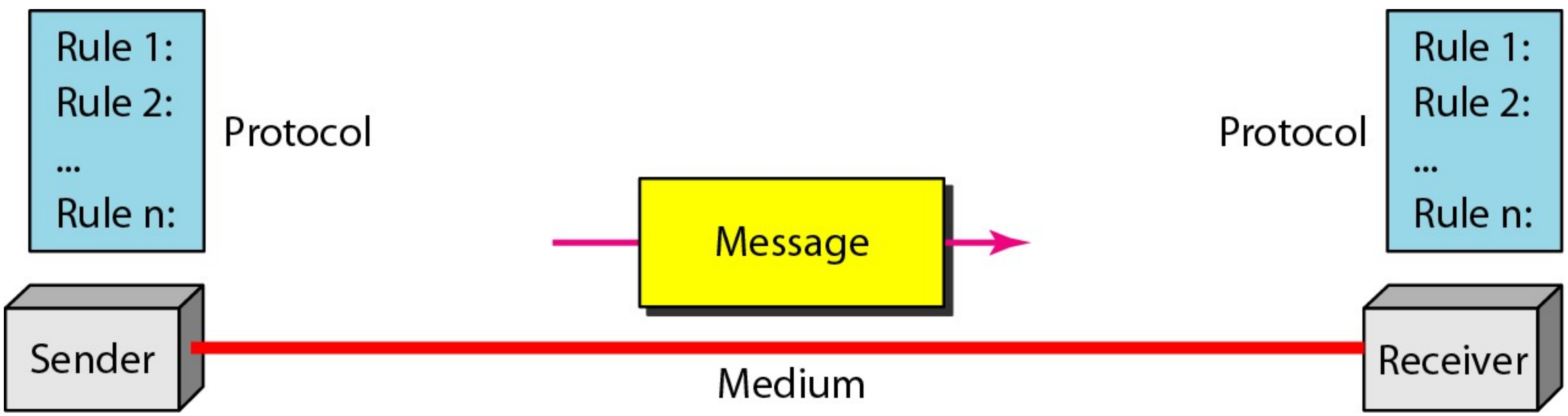
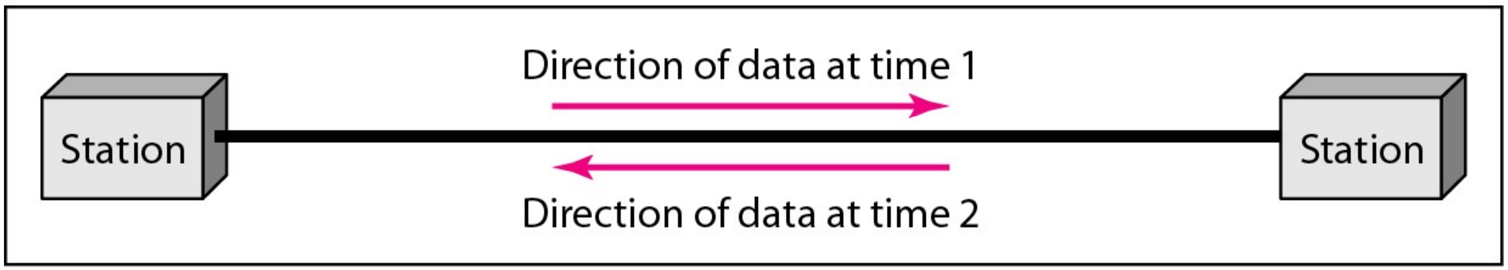
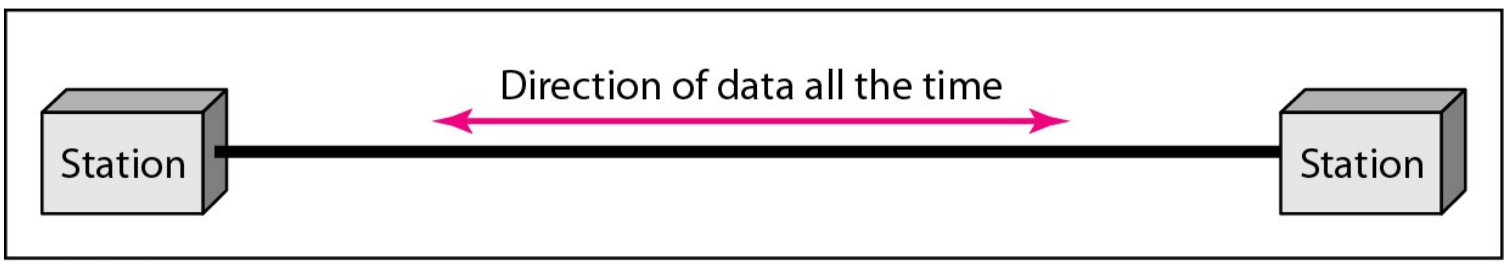


Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*

a. Simplex

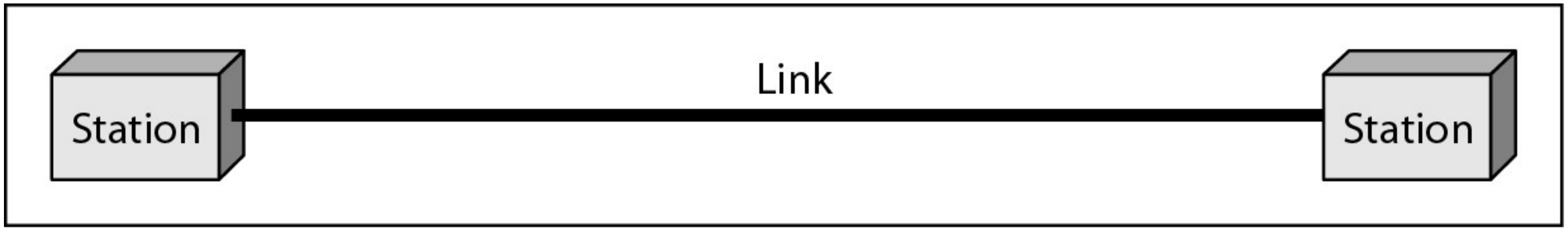


b. Half-duplex

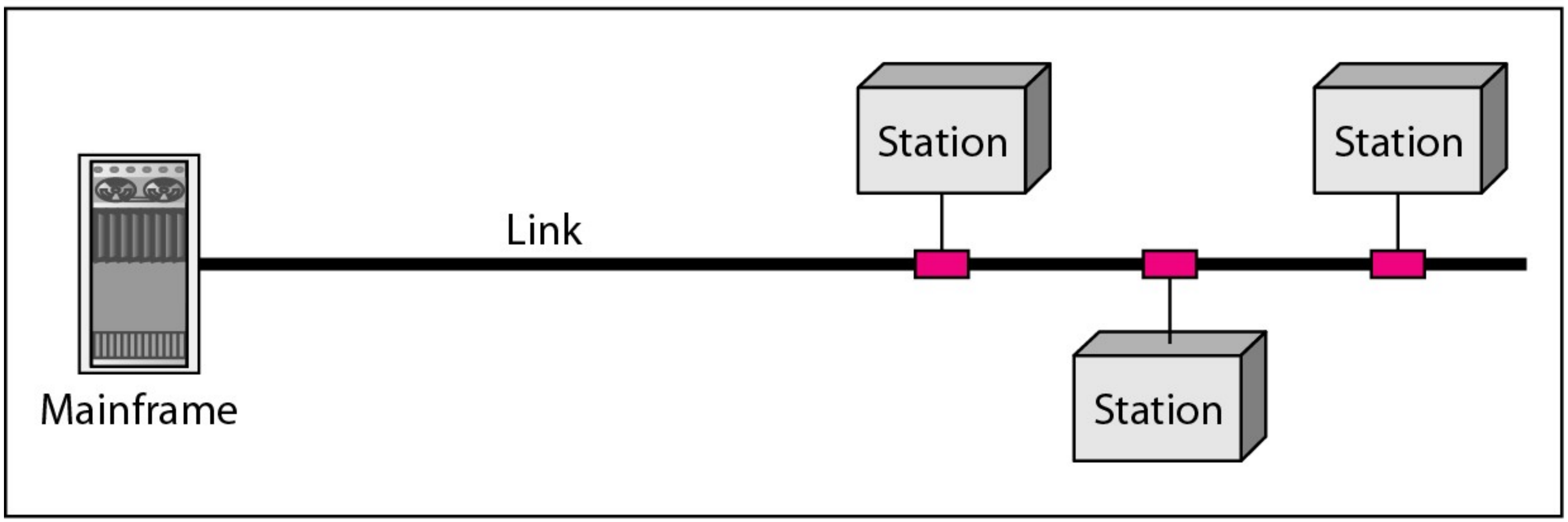


c. Full-duplex

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point



b. Multipoint

Evolution of Network - A brief history

In mid- 1960's - Advanced Research Project Agency(ARPA) in DoD

In 1967, Association of Computing Machinery (ACM) presented idea for ARPANET

In 1969, ARPANET connected via IMPs to form a network

Software called NCP provides communication between the hosts.

In 1972, APRANET collaborated with Internetting Projects to achieve end-to-end delivery of packets

Network Topology

- Computer network topology is the way various components of a network are arranged.
- It defines the layout, virtual shape or structure of network, not only physically but also logically.
- The way in which different systems and nodes are connected and communicate with each other is determined by topology of the network

Topology can be physical or logical

- ***Physical topology*** is the physical layout of nodes, workstations and cables in the network.
- ***Logical topology*** is the way information flows between different components.

Figure 1.4 *Categories of topology*

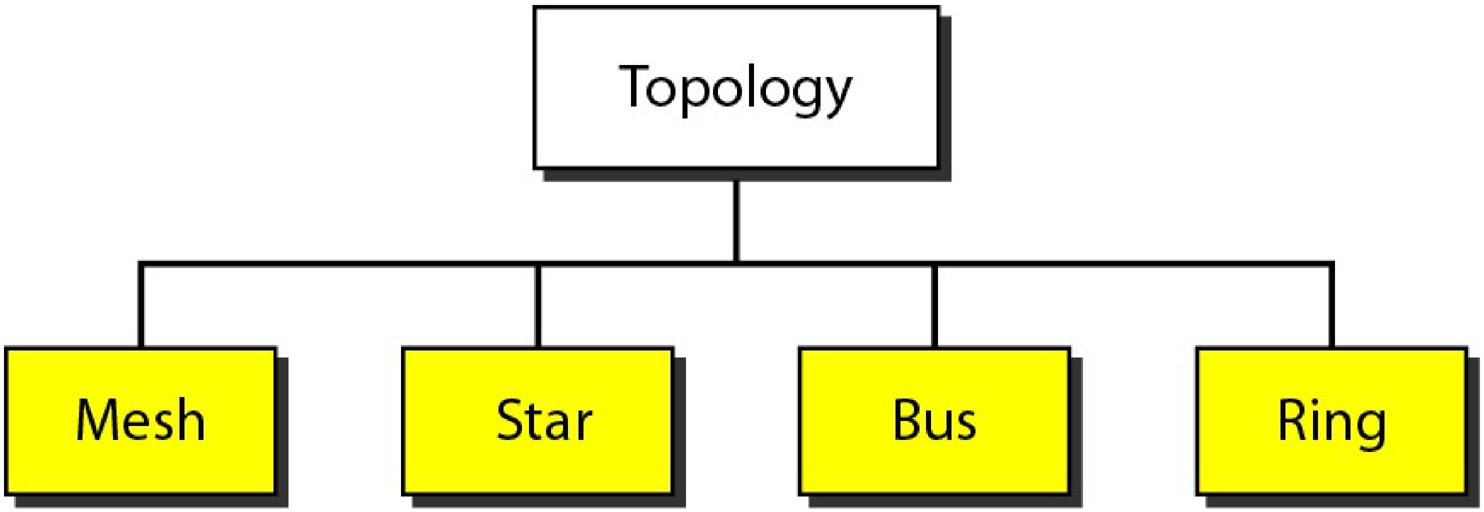
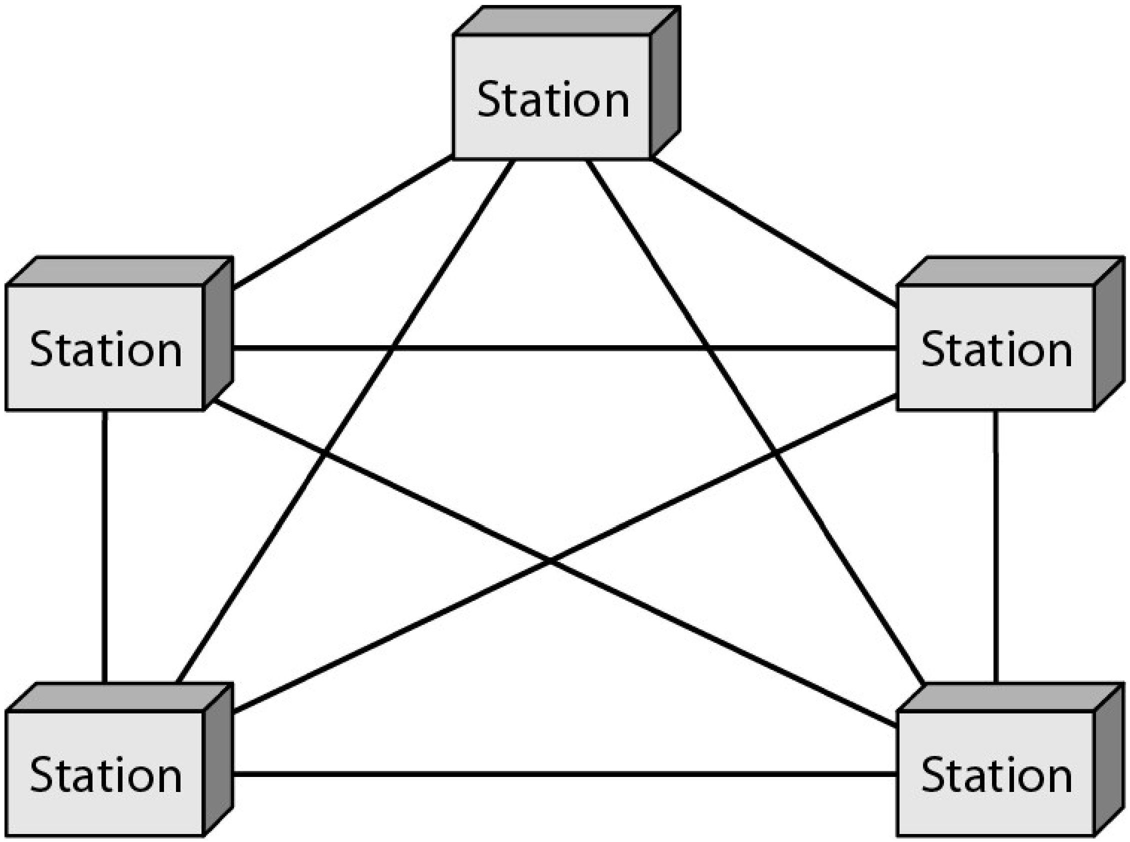


Figure 1.5 *A fully connected mesh topology (five devices)*



- Every device has a dedicated point-to-point link to every other device.
- Each node must be connected to n-1 nodes.
- Fully connected mesh network with n nodes, we need n(n-1) physical links.
- Communication as duplex mode.
- Mesh topology needs n(n-1)/2 duplex mode links
- Every device on network must have n-1 I/O ports

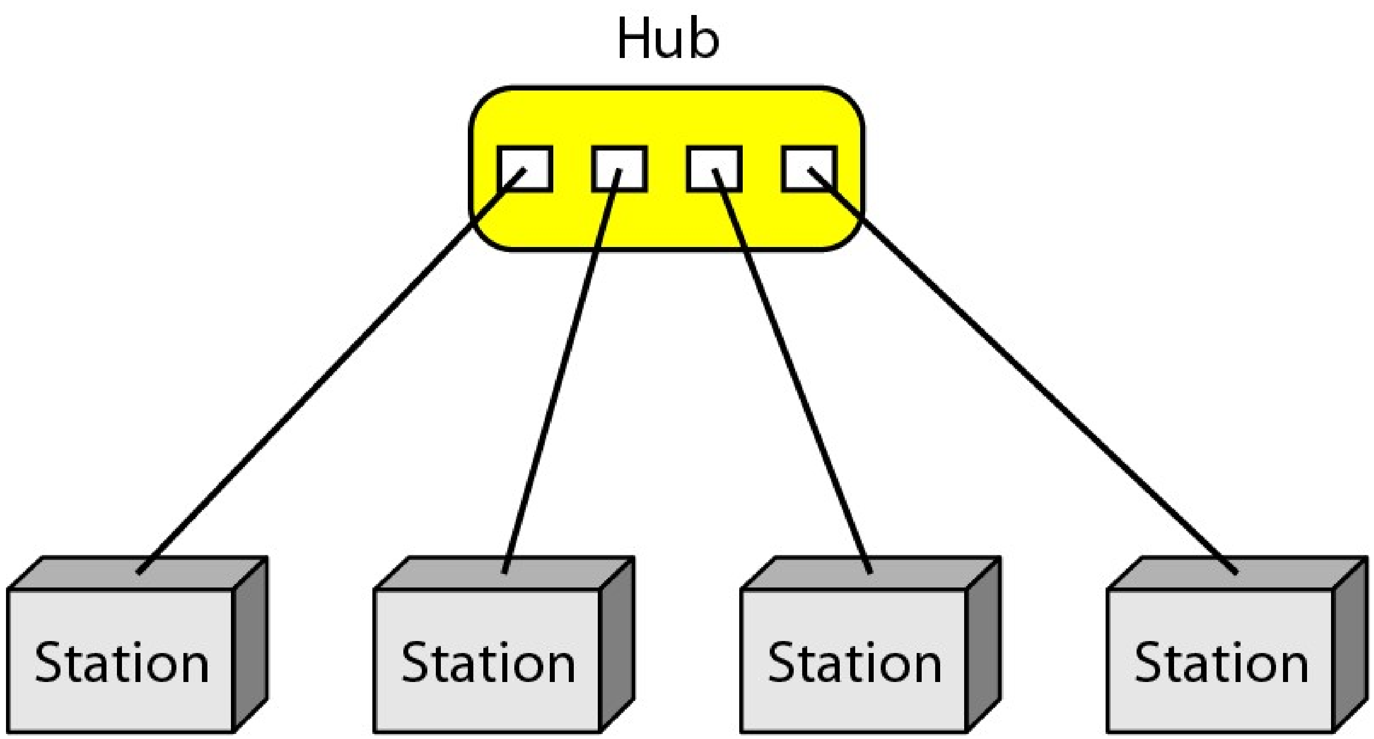
Advantages:

- Use of dedicated links between station
- Eliminated traffic problems
- It is robust
- Privacy and security
- Easy fault identification and fault isolation

Disadvantages:

- Amount of cabling and the number of I/O ports
- Installation and reconnection are difficult
- Bulk of wiring accommodate more available space
- Hardware required to connect each link can be expensive

Figure 1.6 *A star topology connecting four stations*



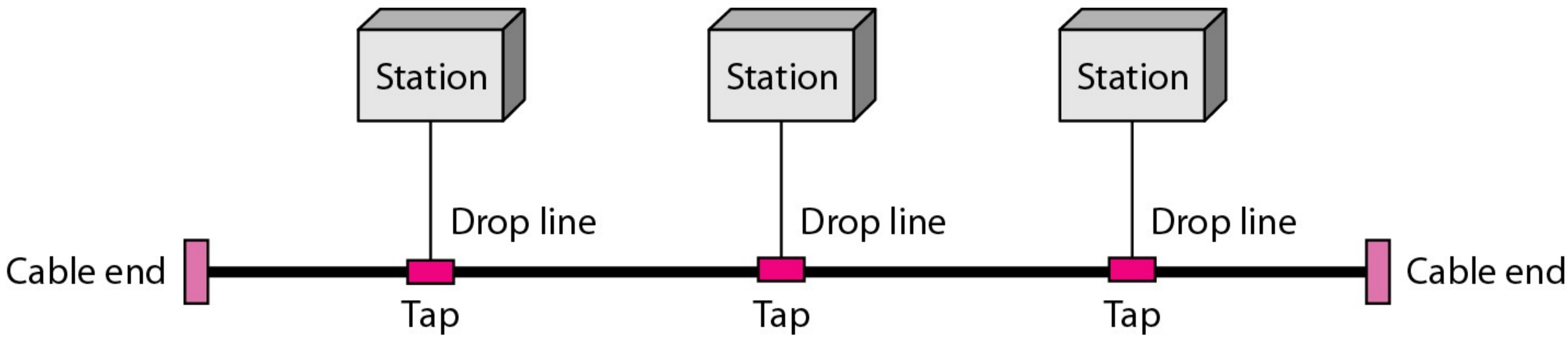
Advantages:

- Less expensive than mesh topology
- Easy installation and reconfigure
- Robustness
- Easy fault identification and isolation

Disadvantages:

- Dependency of whole topology on one single point
- More cabling is required than in some other topologies(bus and ring)

Figure 1.7 *A bus topology connecting three stations*



- Bus topology is multi-point
- One long acts as a backbone to link all devices
- Nodes are connected to the bus cable by drop lines and taps.

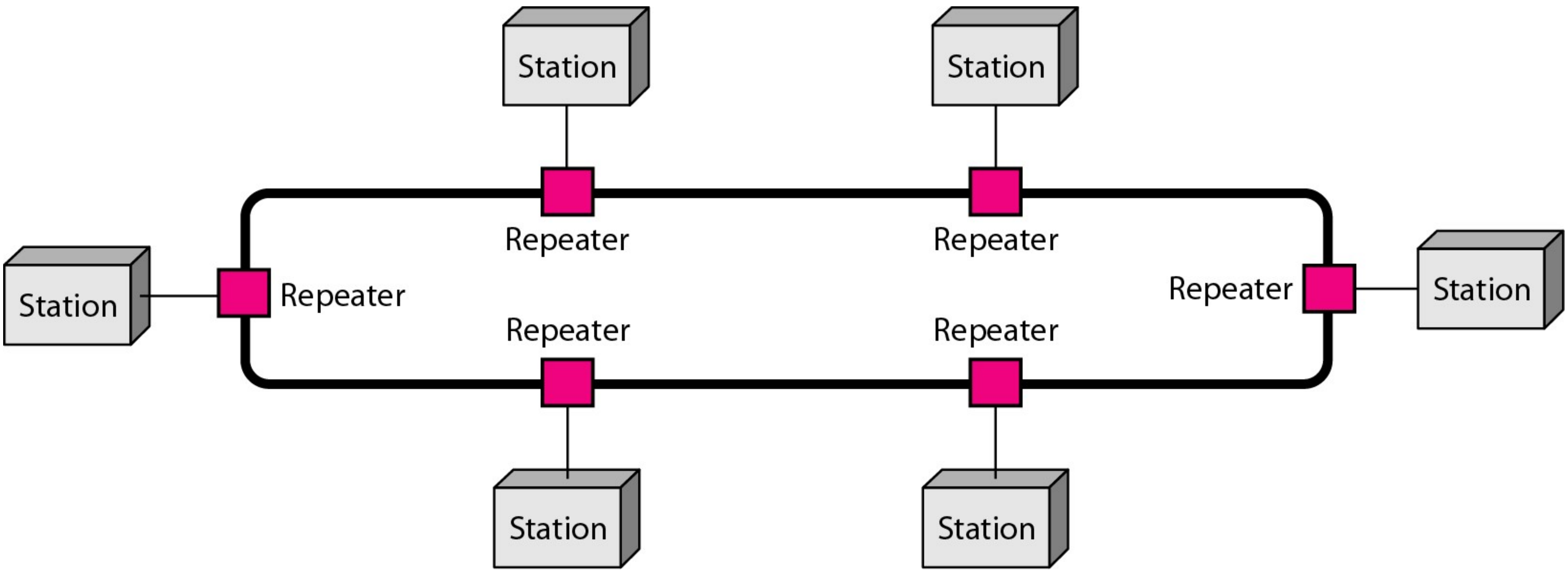
Advantages:

- Ease of installation
- Less cabling than mesh and star topologies

Disadvantages:

- Difficult reconnection and fault isolation
- Difficult to add new devices

Figure 1.8 *A ring topology connecting six stations*



Advantages:

- Ease of installation and reconfigure
- Add or delete a device requires changing only 2 connections
- Fault identification is simplified

Disadvantages:

- Unidirectional traffic
- Break the ring can disable the entire network

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*